

Cloud Architecture for Horizon Stay

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Overview

Horizon Stay is a modern hospitality organization focused on creating a superior guest experience through advanced digital technologies. The company uses a cloud-based **Lakehouse architecture** to manage large volumes of both structured and unstructured data from sources such as mobile apps, web platforms, POS systems, and reservation systems.

The architecture enables real-time data ingestion, centralized analysis, and personalized guest engagement. This scalable and intelligent system is tailored to support operational automation and strategic decision-making.

Purpose & Mission

Purpose:

To integrate and analyze real-time data from multiple hospitality sources, enabling Horizon Stay to deliver personalized guest experiences, optimize operations through automation, and support strategic decision-making using a secure, scalable cloud-based architecture.

Mission:

To provide seamless, intelligent, and tailored hospitality experiences using innovation and data-driven services powered by cloud technology.

Objectives

- **Data-Driven Optimization:** Leverage insights for operational efficiency and smarter decision-making.
- **AI Decision Intelligence:** Use machine learning for dynamic pricing, guest behavior trends, and demand forecasting.
- **Personalized Guest Experience:** Deliver customized services, alerts, and recommendations.
- **Scalability and Performance:** Support growth across various locations without impacting performance.
- **Security and Compliance:** Ensure safe handling of customer data with regulatory compliance.

Design and Discovery Phase

Understanding Hospitality Requirements

- Handle structured and unstructured data from various digital touchpoints.
- Integrate with mobile and web apps, and POS systems.
- Maintain data security and compliance (e.g., GDPR).

Key Challenges

- **Data Silos:** Scattered and uncoordinated data systems.
- **Inconsistent Formats:** Difficulty in data normalization.
- **Integration Complexity:** Diverse data sources and tools.

Opportunities

- **Unified Lakehouse Architecture:** Enables centralized processing.
 - **Real-Time Engagement:** Live tracking and response to guest activities.
 - **Predictive Intelligence:** Forecast trends, needs, and behaviors.
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Data Sources and Sinks

Source	Data Type	Flow	Sink	Outcome
Social media	Text, Images, Metadata	Streaming	AI-Driven Marketing Engine	Campaign optimization, sentiment trends
Geo-location	GPS Coordinates	Batch	Logistics Platform	Guest path tracking, navigation
Web/Mobile Applications	Clickstream, Bookings	Batch	Notification Engine	Personalized offers, real-time updates
POS Systems	Financial Transactions	Batch	Revenue & Financial Insights	Guest spending behavior, financial KPIs
Reservation Systems	Booking Info	Batch	Smart Inventory System	Room allocation, demand prediction

Key Deliverables

- **Logistics Planning:** Resource scheduling and GPS-based guest guidance
 - **AI-Driven Marketing Intelligence:** Personalized promotions, sentiment insights
 - **Smart Inventory:** Real-time room availability, demand forecasting
 - **Revenue & Finance Insights:** KPI dashboards, guest expenditure trends
 - **Guest Notifications:** Real-time, event-triggered alerts and offers
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Cloud Architecture

Bronze Layer – Ingest Data

- Raw data is collected from all systems (CSV, JSON, Parquet)
- Tools used: **Azure Data Lake Gen2, Event Hub, Data Factory**

Silver Layer – Curate Data

- Cleansing and enrichment using **Azure Synapse Analytics**
- Standardization, normalization, and transformation of data
- Trusted datasets are generated for business use

Gold Layer – Aggregate Data

- Final analytics-ready datasets
- Examples: Revenue reports, satisfaction trends, inventory forecasts

Phased Architecture

1. Initial Phase:

- Set up data lake, raw ingestion via pipelines
- Test connectivity and input streams

2. Refined Phase:

- Transform and enrich data
- Cleanse inconsistent formats, unify schema

3. Final Phase:

- Implement BI dashboards, real-time alerts, and scalable outputs
 - Optimize for performance and automation
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Pipeline Strategy and Fault Handling

Pipeline Orchestration:

- A **master pipeline** manages sub-pipelines using event-based triggers
- Each subsequent pipeline runs based on the success/failure of the previous one

Failure Handling:

- **Retry Attempts:** Automatically retries up to 3 times
- **Retry Interval:** 30-minute pause between retries
- **Error Logging:** Failures are logged and flagged for manual intervention

Conclusion and Future Scope

The Horizon Stay cloud architecture empowers the hospitality sector with a powerful data-driven platform. It supports real-time personalization, automated operations, and predictive insights that help elevate the guest experience. The Lakehouse model is scalable, secure, and flexible—paving the way for future integration with advanced AI, multi-property data systems, and deeper analytics models.